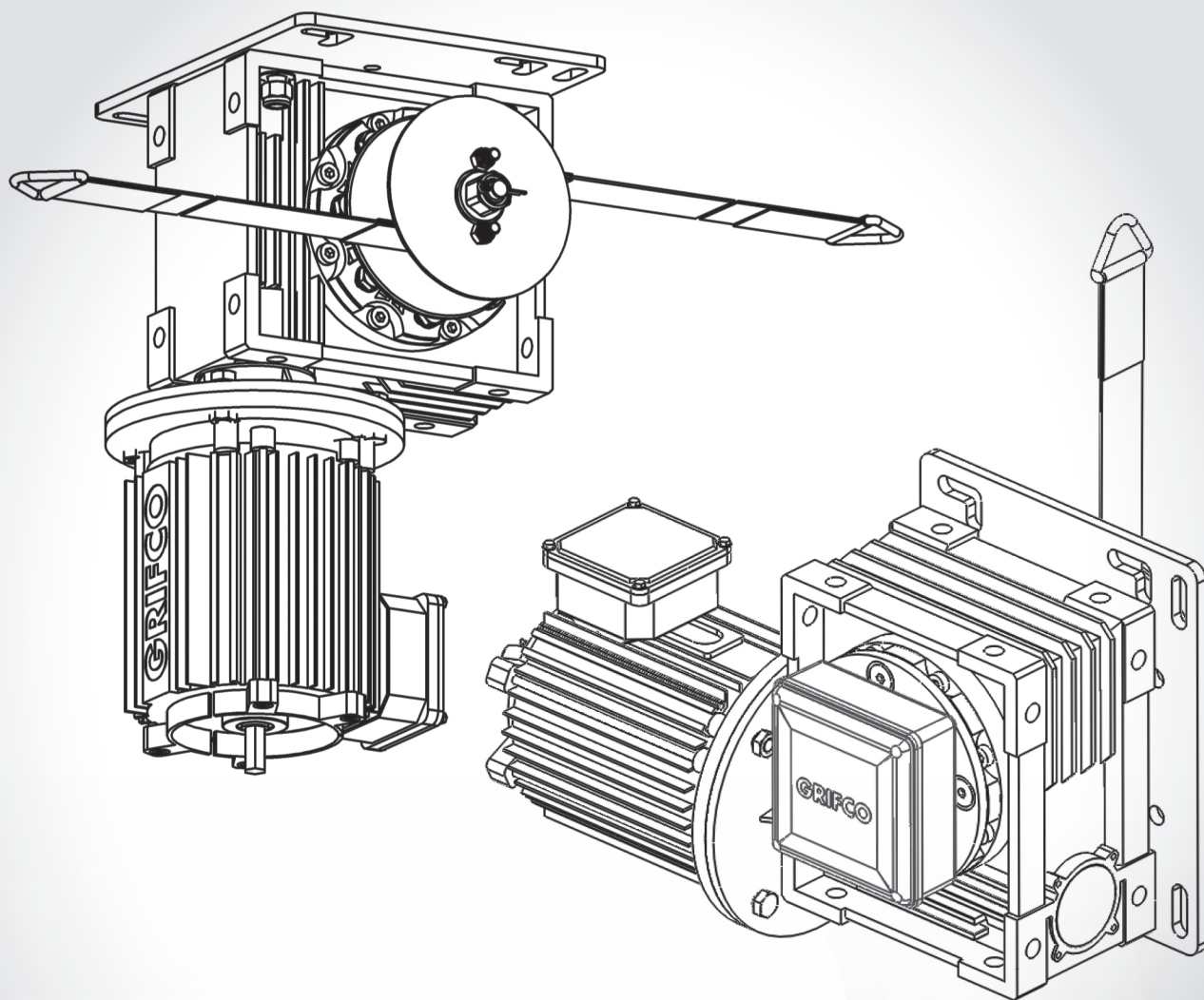


W400B Universal Belt Winch INSTALLATION MANUAL



W400B Universal Belt Winch

Installation and Users Manual for
W400B3 and W400B1 Universal Belt Winch

Chamberlain Australia Pty Ltd
Unit 1, 75 Epping Rd
North Ryde NSW 2113
Ph: 1800 GRIFCO
www.grifco.com.au

Chamberlain New Zealand Ltd
17B Arrenway Drive, Rosedale 0623
Auckland
Ph: 0800 653 667
www.grifco.co.nz

Ref: D-M-PSA-028_B

02-04-2025



GRIFCO®

SAFETY INSTRUCTIONS



WARNING: THESE ARE IMPORTANT SAFETY INSTRUCTIONS.

FOLLOW ALL INSTRUCTIONS AS INCORRECT INSTALLATION CAN LEAD TO SEVERE INJURY OR DEATH

SAVE these instructions

This winch has been designed and tested to offer safe service provided it is installed, operated, maintained and tested in strict accordance with the instructions and warnings contained in this manual.



WARNING

Mechanical

When you see this Safety Symbol and Signal Words on the following pages, they will alert you to the possibility of serious injury or death if you do not comply with the warnings that accompany them. The hazard may come from something mechanical.



WARNING

Electrical

When you see this Safety Symbol and Signal Words on the following pages, they will alert you to the possibility of serious injury or death if you do not comply with the warnings that accompany them. The hazard may come from electric shock.

DANGER

The use of the word DANGER calls attention to high risks associated with this product..



The winch must be installed in accordance with the installation codes and regulations applicable in the country where it is installed- for Australia and New Zealand electrical installations must comply with AS/NZS 3000.



An adequate and safe power supply is essential for safety. A competent and qualified electrician must undertake all electrical wiring. A main isolator should be installed with the winch and is essential for safety.



Before you do any repair, or maintenance works, always disconnect the electricity supply.



Installation, maintenance and repairs should be conducted by technically trained and competent persons only.



Intended Use: This winch is designed solely for use inside livestock buildings to hoist feeder lines and other similar systems. Use of the winch in any other way is a misuse of the product and may endanger safety and health.



DANGER! Structural Safety: The weight of this winch and the raising and lowering of the associated feeding system in a livestock building imposes additional load on the building. A structural engineer should verify the building's load capacity prior to installation of this winch.



Practice Safe Maintenance: Keep your winch in proper working condition by maintaining it in accordance with these instructions. Unauthorized modifications to the winch may impair the function and/or safety and affect its life.



Ensure you wear personal protection equipment (gloves, dust masks). Never use your hands at dangerous locations (eg drive units) before you have completely switched off the electricity supply and made sure that nobody can switch it on without your knowledge.



DANGER: A wire cable under tension can cause very serious injuries when released or if it snaps. The cable used in this installation may break at any time so appropriate precautions should be followed.



DANGER:

- Never allow unauthorized persons to enter the building in your absence or during servicing of the equipment.
- Be careful when lowering or winching up the feeding lines/circuits :
 - i. stop operation immediately at the slightest abnormality in behaviour.
 - ii. never stay underneath the winch or load when lowering or winching up the lines/circuits.
- If the drive stalls : immediately switch off the system. Carefully read the trouble shooting guide and strictly follow instructions. Contact a technically trained person.

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SPECIFICATIONS

W400B General Specification		
Belt size	0 to 3.5 Meter	3.5 to 4.0 Meter
Rated Load (double belt)	2 x 200Kg	2 x 175Kg
Rated Load (single belt)	1 x 250Kg	1 x 200Kg
Max Travel	2.8m	3.8m
Lifting Belt*	50mm x 1.6mm Polyester	
Lift Speed	2.6-4.8m/min	
Drum RPM	14	
Limits	2 x N/C Micro Switch	
Unit Weight	28Kg	
Protection	IP55	
Working Life	20yr / 5000 cycles	

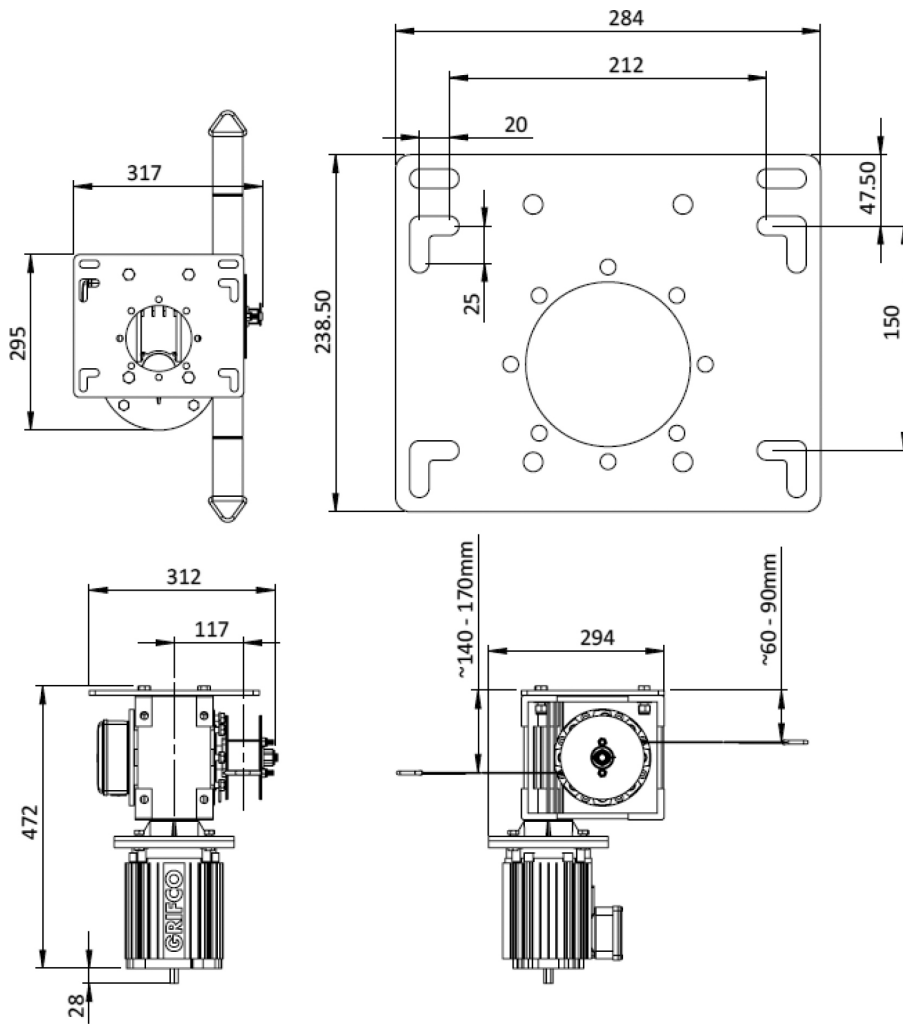
W400B3 Power Specification	
KW	0.75
Full Load Current	2.3A
Volts (AC)	415
Phase	3

W400B1 Power Specification	
KW	0.75
Full Load Current	4.5A
Volts (AC)	240
Phase	1

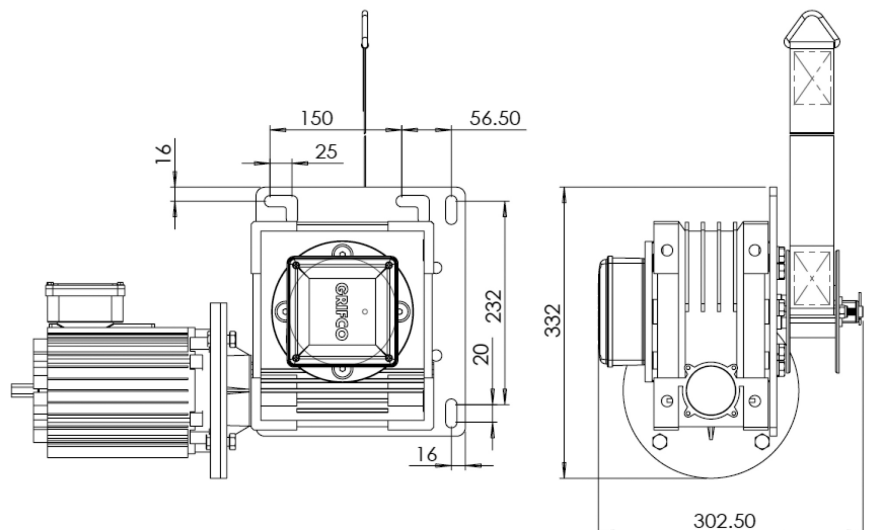
* Only Grifco lifting belts may be used with this product

SPECIFICATIONS

CEILING MOUNT CONFIGURATION—Double pull belt (standard, as supplied)



FLAG MOUNT CONFIGURATION—Single pull belt (optional)



INTRODUCTION

Congratulations on your purchase of the Grifco® Universal Belt Winch W400B. This winch stands alone in the market as the fastest and most robust winch of its kind. The installation of a Grifco Universal Belt Winch ensures extreme longevity and trouble free operation.

PLANNING

The W400B is designed to be installed inside or outside livestock buildings to hoist drinker lines, ventilation curtains or shutters and many other similar applications.

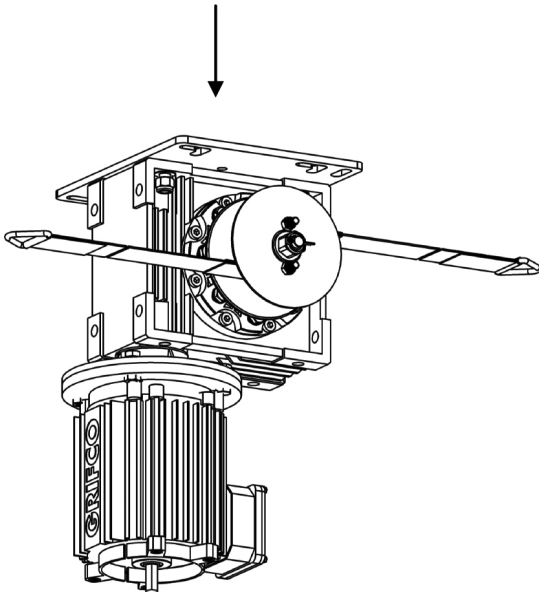
You will find that throughout this manual, there are two mounting configurations referred to, i.e. Ceiling Mount, and Flag Mount. Refer to “Configuration” below to demonstrate the difference. The winch is supplied standard as Ceiling Mount.

The W400B features a single or three phase motor coupled to a drive system comprising two cam activated limit switches to allow stopping at the fully lowered and raised positions.

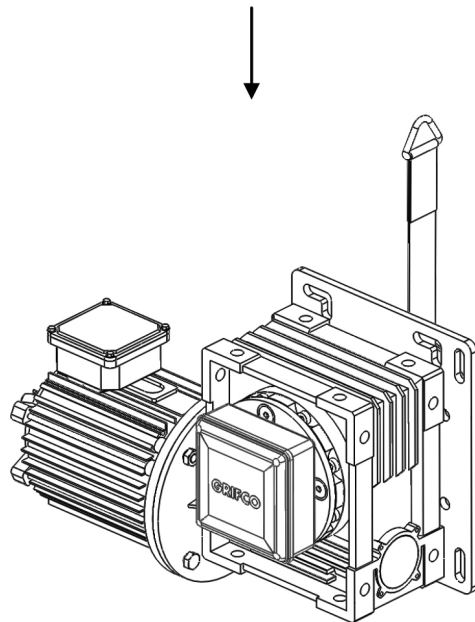
Suitable controls will be required for the W400B to switch the motor direction and accept the limits switch inputs. Grifco manufactures both single and multi-line control stations for the W400B.

CONFIGURATION

CEILING MOUNT CONFIGURATION
Double pull belt (as supplied)



FLAG MOUNT CONFIGURATION
Single pull belt (alternate)



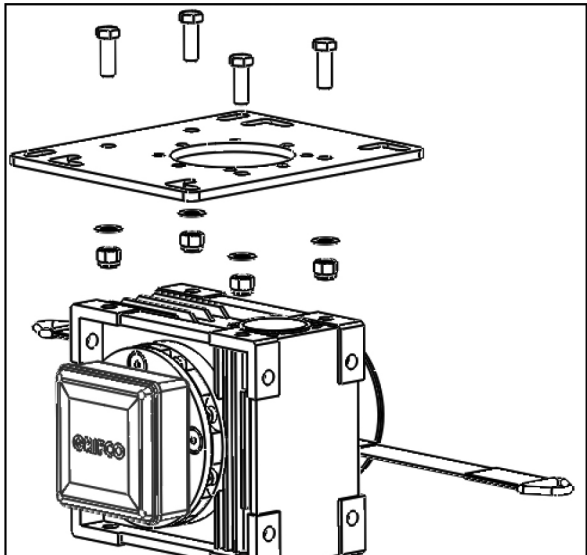
The next page demonstrates the process of converting your winch to Flag Mount

INTRODUCTION

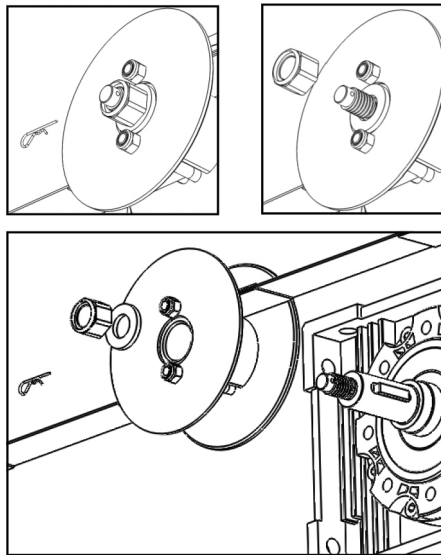
CONVERSION

To convert the Ceiling Mount configuration to Flag Mount, follow the steps below:

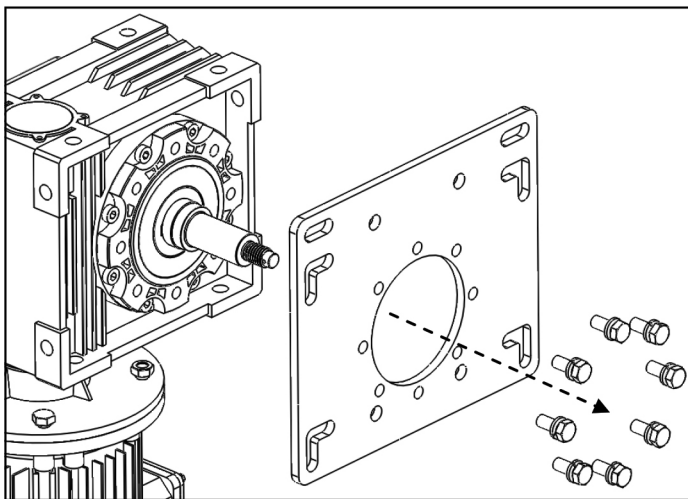
Step 1: Remove the 4 x M12 bolts, washers and nuts.



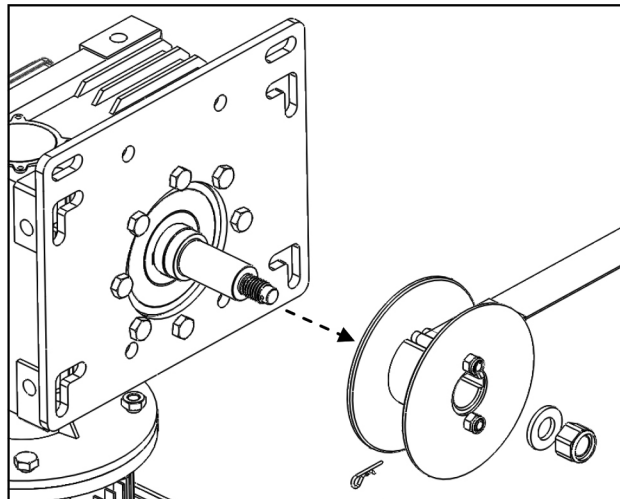
Step 2: Remove the “R” clip, nut and drum assembly



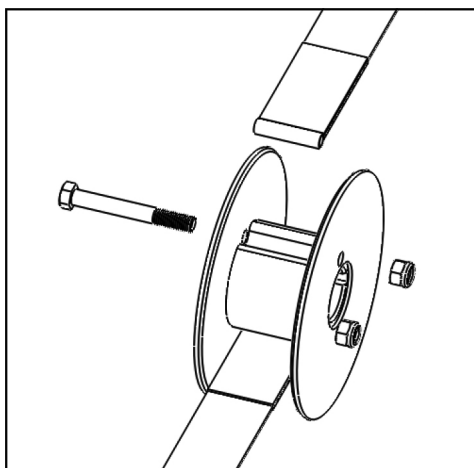
Step 3: Fit the mounting plate in a flag orientation as shown, using the 8 x M10 fasteners and washers provided (tighten to 30Nm).



Step 4: Fit the drum over the shaft and key, ensuring it is secured tightly with the nut, and “R” clip in place.



Belt Removal: To remove or change a belt, simply loosen the relevant M8 bolt and nut, and slide out to allow the belt to come free. Reassemble the M8 bolt and nut and tighten firmly.

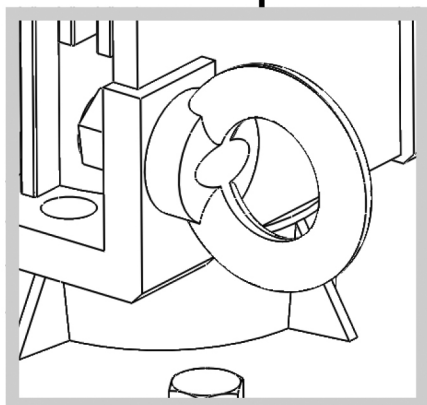
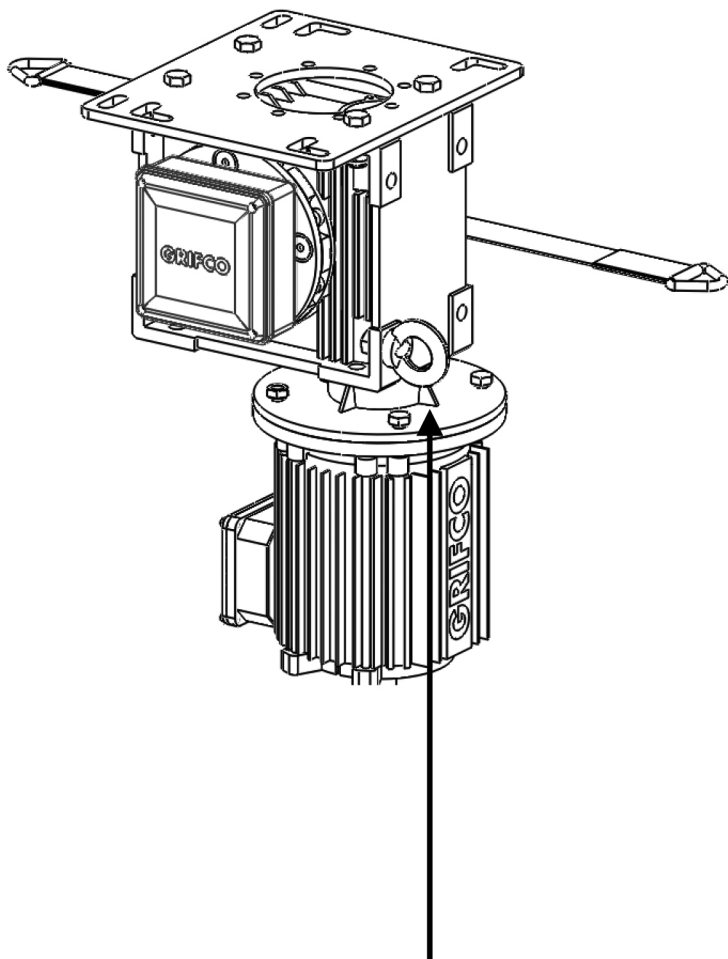


HANDLING

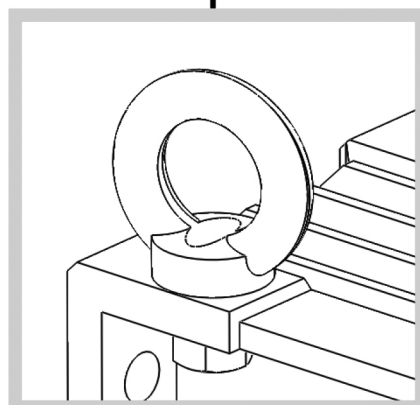
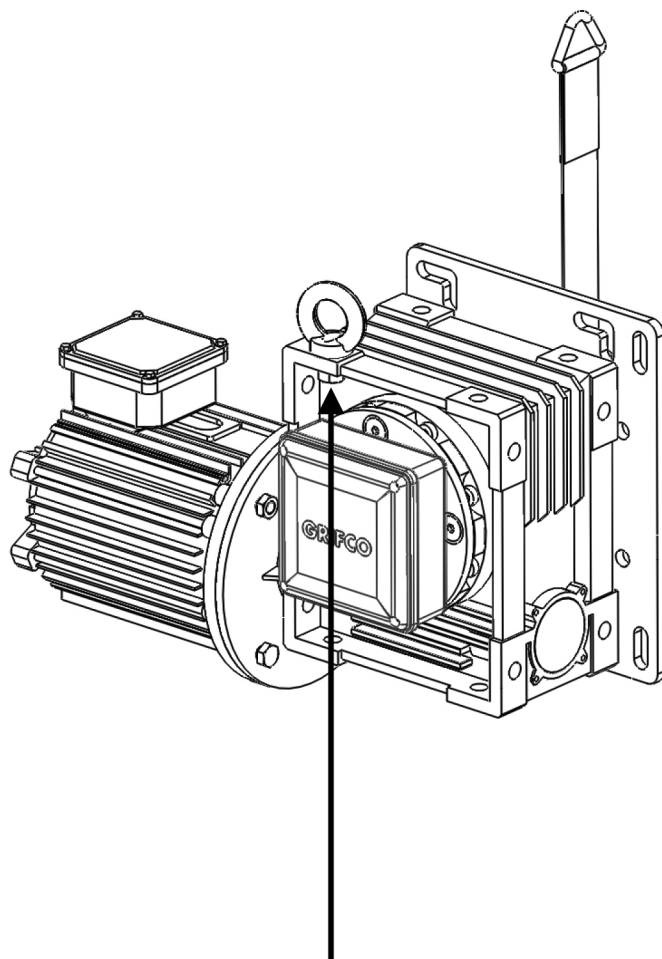
Ensure you wear personal protection equipment (proper clothing, gloves, dust masks). Never use your hands at dangerous locations (eg sharp edges etc).

In order to safely lift the W400B into the final mounting position, use an approved M10 or M12 lifting eye & nut as shown below:

Ceiling Mount



Flag Mount



CAUTION

When using the above lifting method, be aware that the winch will not hang in a level orientation (as pictured)

MOUNTING



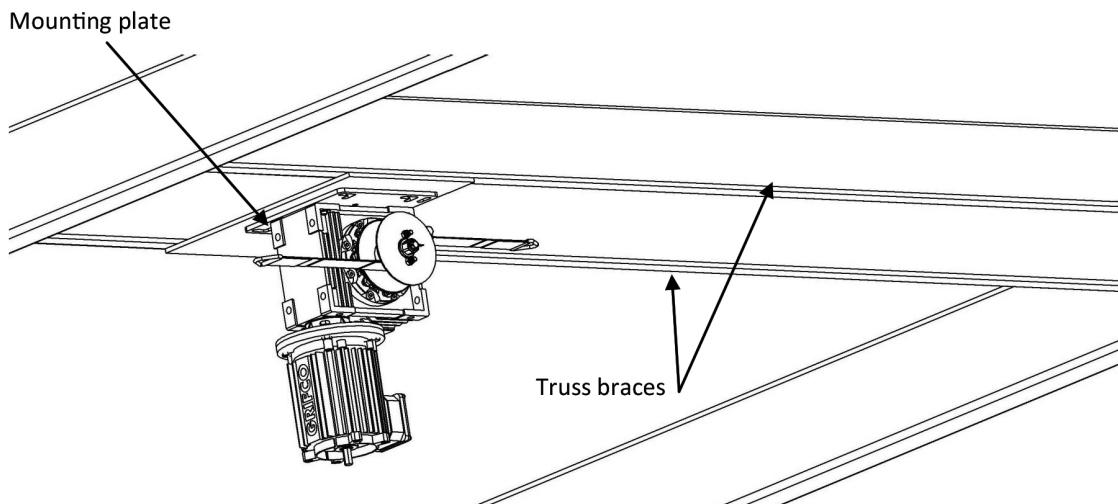
WARNING

Electrical

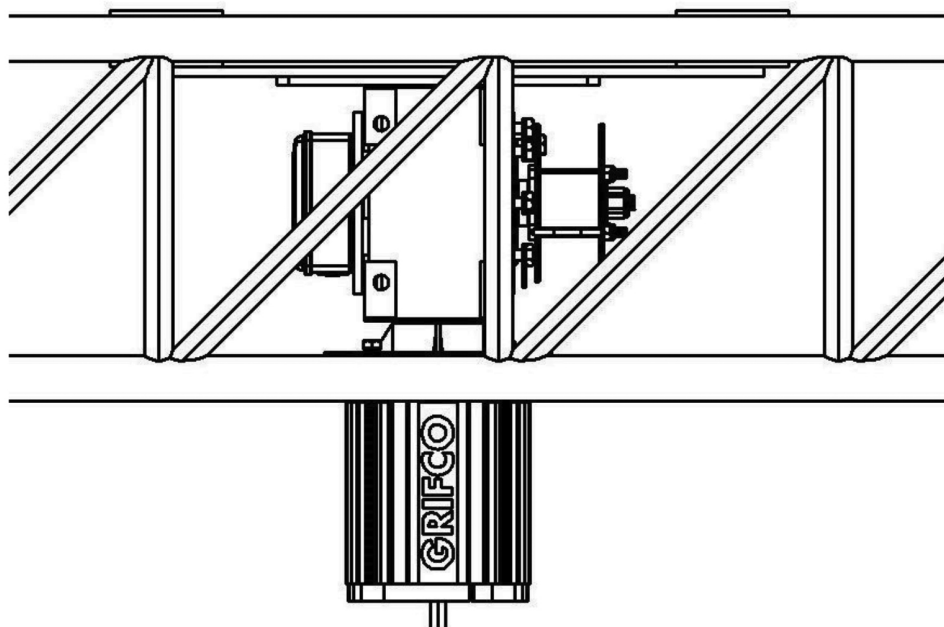
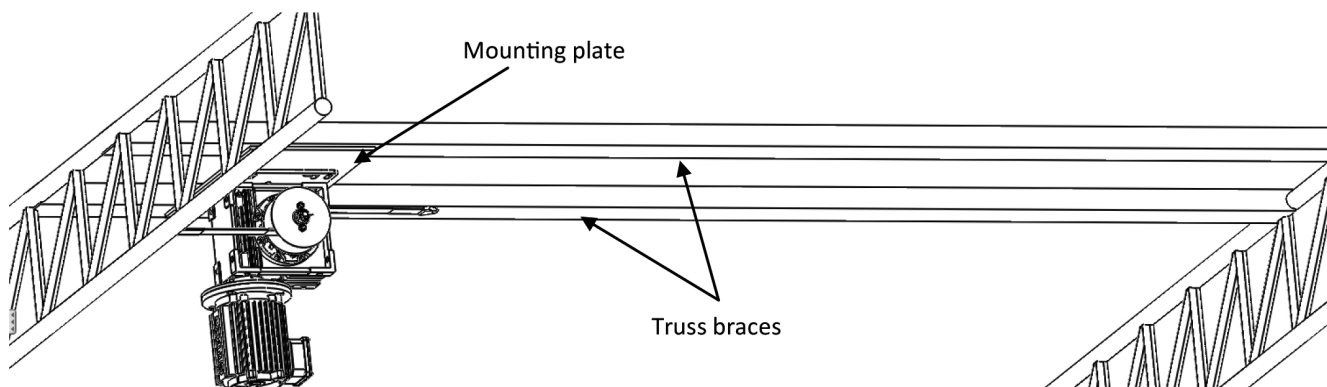
An electrician must disconnect electric power to the winch and control before removing or opening covers.

The W400B is designed to be mounted to a concealed or web truss structure in livestock buildings. The roof truss structure must be engineered to ensure mounting of the winch and the relevant applied loads will be adequately supported. Refer to page 4 for mounting bolt hole spacing and size for fabrication of a suitable mounting structure. Refer below for mounting examples.

Concealed Truss Mount



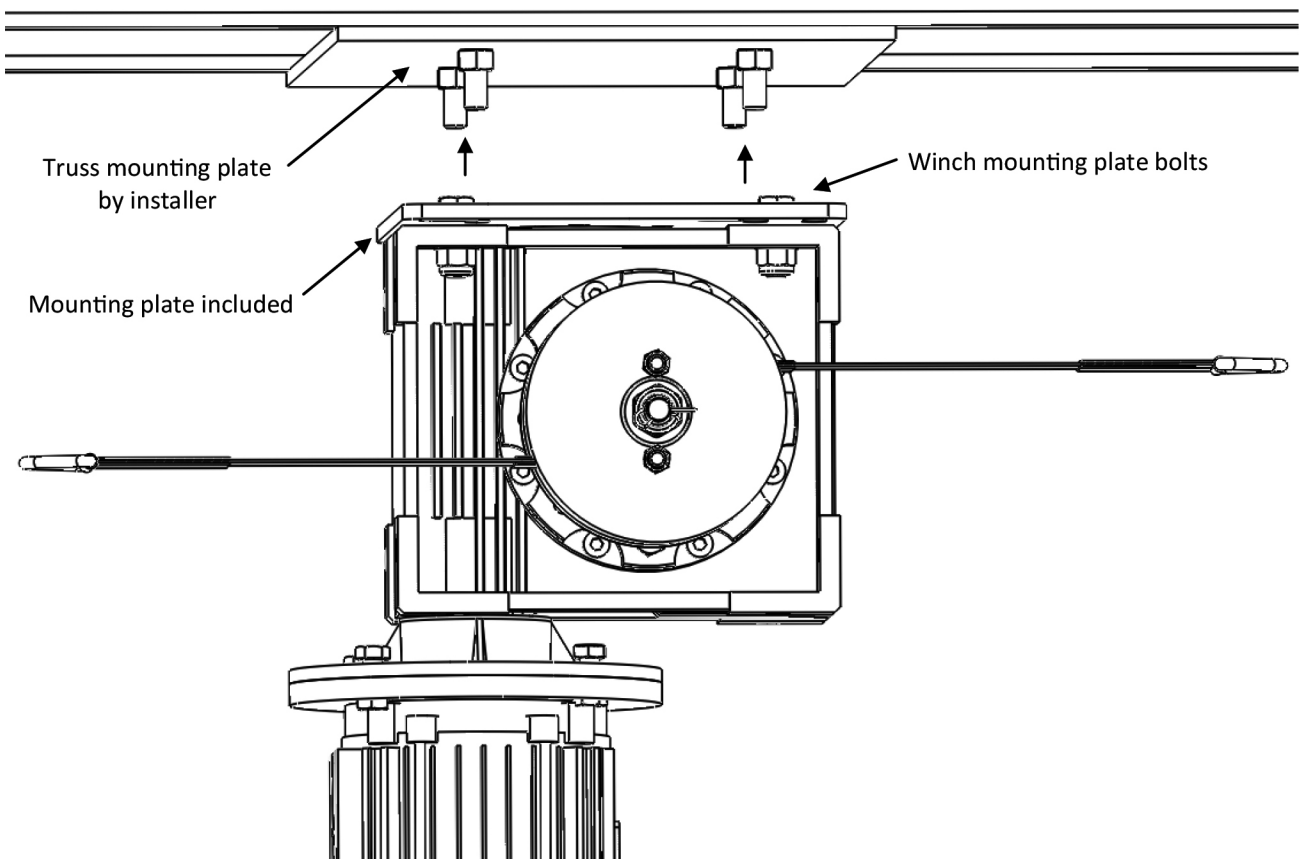
Web Truss Mount



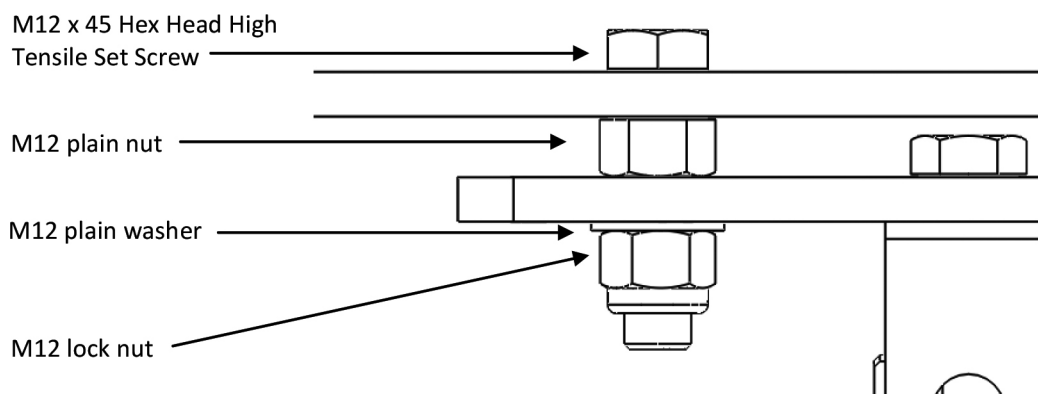
MOUNTING

CEILING TYPE

Mount the W400B using 4 x M12 High Tensile (Grade 8.8) bolts and nuts to secure the winch in place as shown below. Ensure spring washers are used and a tightening torque of 40-45Nm. As seen below, by pre-fitting the 4 x M12 x 45 long high tensile hexagon head fasteners, with nuts (to also act as a spacer to clear the **winch mounting plate bolts**) you will have 4 x studs to allow the winch mounting plate to slide over (Note: Fasteners are not supplied). Weld fixing may be suitable if carried out in accordance with proper structural advice and by a suitably qualified person.



When bolt fixing the W400B as shown above, ensure a fastener combination is used as below. Fasteners must be M12 x 45 hexagon head high tensile set screws, including 1 x flat washer, 1 x plain nut, and 1 x lock nut per hole.



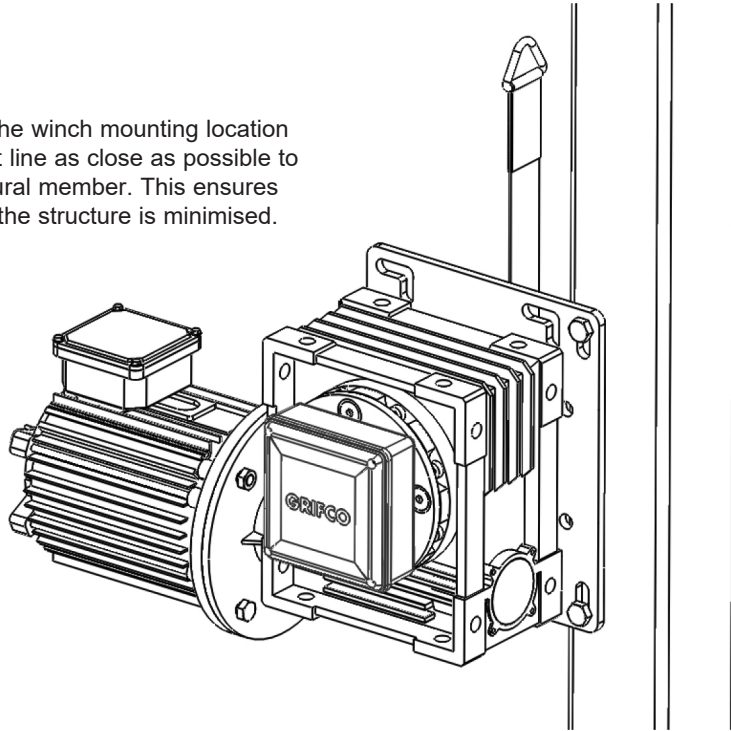
Given the universal design of the W400B mounting system, many mounting variations (different to what is shown in this manual) may be utilised in order to meet your installation requirements. Always ensure the mounting structure used is suitably engineered to withstand the various forces applied during operation in all conditions.

MOUNTING

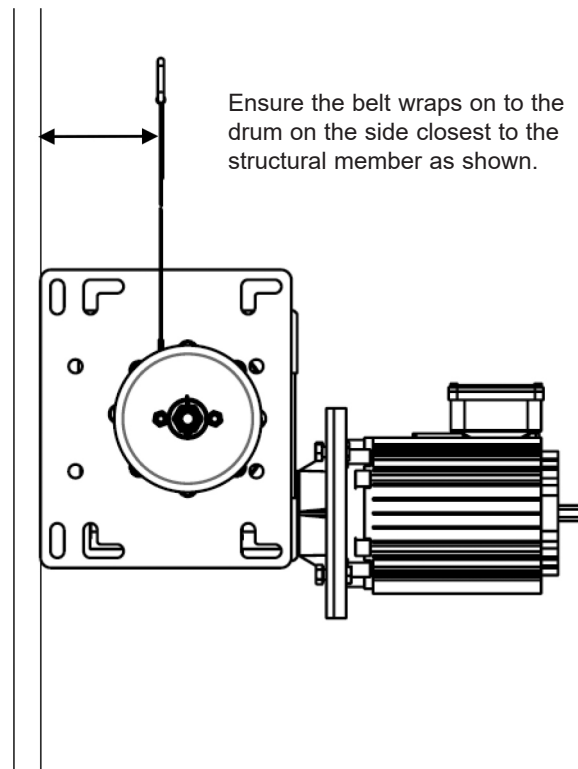
FLAG TYPE

Mount the W400B flag mount type using at least 2 x M12 High Tensile (Grade 8.8) bolts and locking nuts to secure the winch to a suitably engineered vertical or horizontal structural member as shown below. Tighten to a torque of 40-45Nm. (Note: Fasteners are not supplied). Weld fixing may be suitable if carried out in accordance with proper structural advice and by a suitably qualified person.

Note how the winch mounting location sets the belt line as close as possible to the structural member. This ensures stress on the structure is minimised.



Ensure the belt wraps on to the drum on the side closest to the structural member as shown.



ELECTRICAL CONNECTION



WARNING

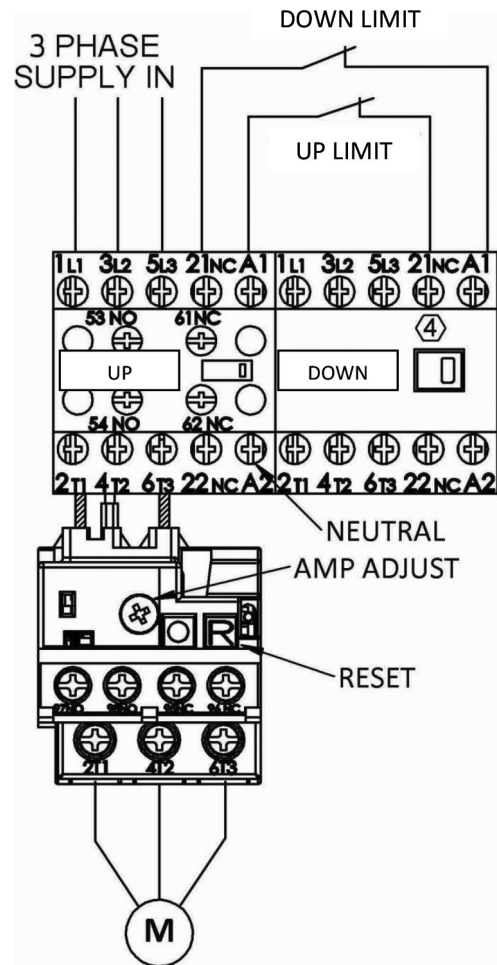
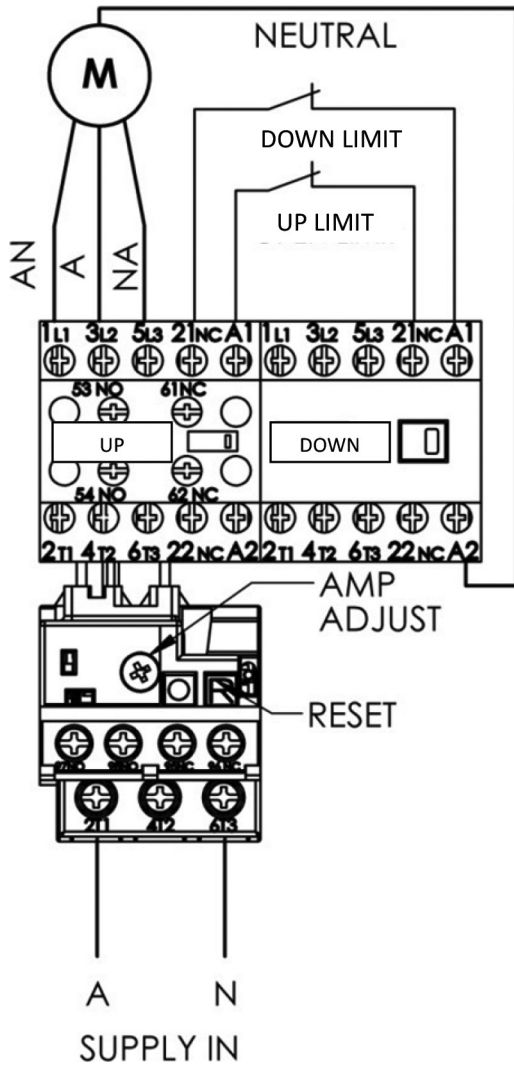
Electrical

The W400B must:

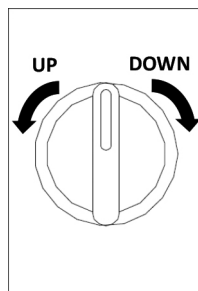
- be connected in accordance with the wiring rules of the country in which it is installed
- not have enclosures left open for extended periods (excess dust will void warranty)

CONTROL PANEL

The W400B is designed to be operated with Grifco reversing starter controls. The internal connection diagram for both single and three phase types are shown below:



Operation of the above control panels is via a simple two position control as shown below. Turn and hold the toggle switch to lift up or lower down as shown:



ELECTRICAL CONNECTION



WARNING

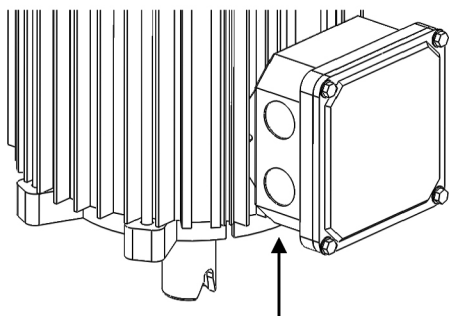
Electrical

The **W400B** must:

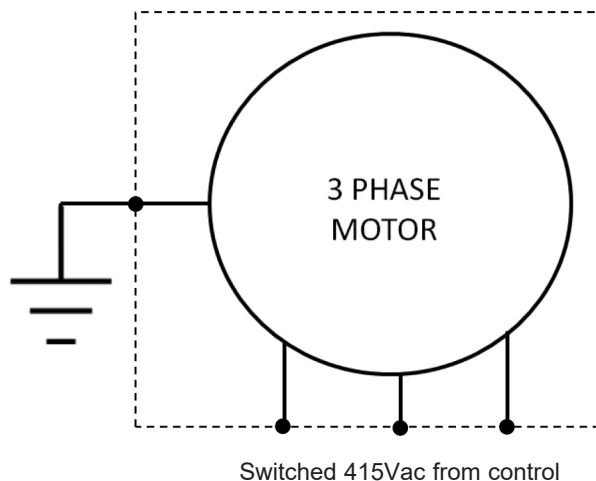
- be connected in accordance with the wiring rules of the country in which it is installed
- not have enclosures left open for extended periods (excess dust will void warranty)

MOTOR CONNECTION

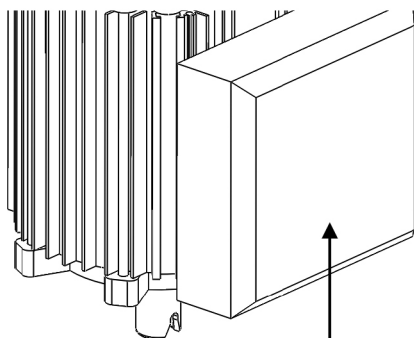
W400B3 Motor Connection



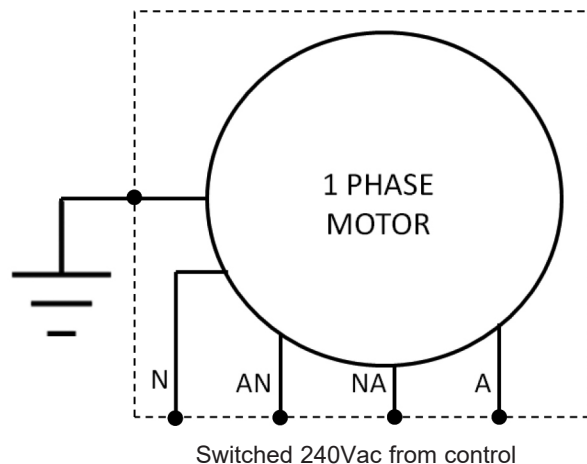
Connect with min. 0.75mm² wire using conduit or gland to meet minimum IP55



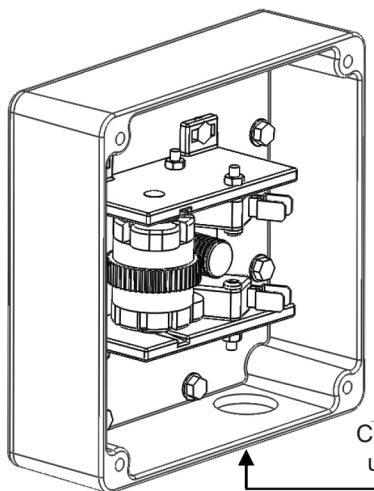
W400B1 Motor Connection



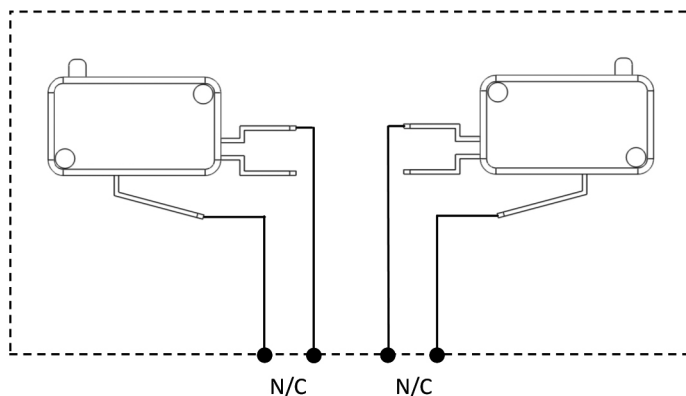
Connect with min. 0.75mm² wire using conduit or gland to meet minimum IP55



Limit Switch Connection



Connect with min. 0.75mm² wire using conduit or gland to meet minimum IP55



SETUP AND ADJUSTMENT

With primary connection of the W400B complete in accordance with relevant control box, it is time to test the operation. Carefully follow the next steps to ensure safe setup:

1. Ensure the winch is properly secured and the lift belts clamped accordingly.
2. Check and make sure both limit cams are positioned away from their relevant micro switches.
3. Secure all covers, and supply mains power.

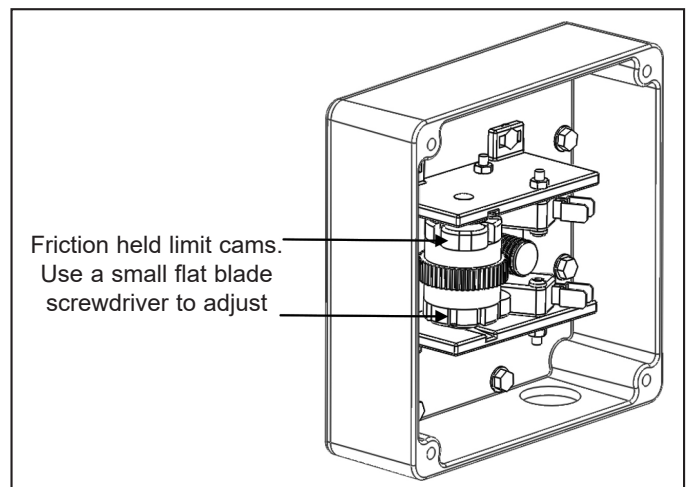
SETTING OPERATIONAL DIRECTION

4. Operate the winch momentarily in one direction to check that the UP switch will lift the load.
5. If incorrect, isolate mains power and proceed as follows: (ref page 13)
 - Swap any two phases on the 3 phase model W400B3
 - Swap AN and NA on the single phase model W400B1
6. Secure covers, supply power to the system, and check for correct operational direction.

SETTING LIMITS

WARNING: Limits are sensitive; a small cam movement may correspond to a large amount winch travel.

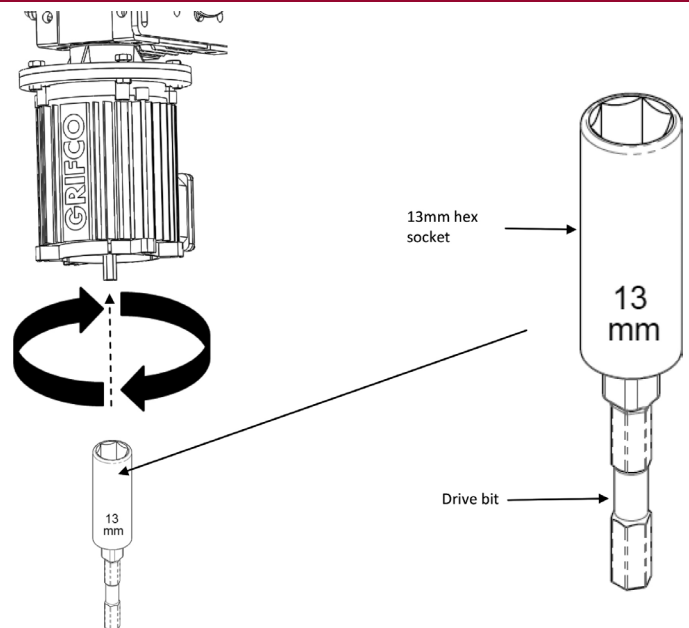
- Position one cam to depress a switch (the other should be well clear of the switch)
 - Note the switch that you have activated
- Supply power and activate in up direction momentarily
 - If the winch hoists up then the limit switch noted above is the lower limit
 - If the winch doesn't move it is the upper limit (momentarily activate in the close direction to confirm this)
- Proceed to adjust each friction held cam and operate the winch to achieve the required set positions.



MANUAL OPERATION

Isolate mains power before using the manual crank feature as shown below. Use a 13mm hex socket with a suitable drive bit (not included) to operate the winch manually. Rotation of the socket will operate the winch in either direction.

NOTE: 13mm hex socket and drive bit are NOT SUPPLIED with the winch



ELECTRICAL OPERATION

In operation the W400B will require the user to hold the controlling switch to raise or lower the load. During operation the belt must lay neatly into drum.

The winch should operate quietly with only the hum of the motor, and perhaps some pulley noise depending on the condition and/or lubrication of the various pulleys connected to the load.

If the winch produces any abnormal noises such as grinding, rattling or squeaking, cease use immediately and isolate the power. Refer to Troubleshooting on page 16, and/or contact a suitably qualified technician to inspect and/or repair.

MAINTENANCE



WARNING

Electrical

DANGER! Power MUST be turned off before servicing or adjusting the winch. Isolate the supply when cleaning! Any works requiring access to electrical connections or junction boxes must be carried out by a qualified electrician!

The W400B is built for extreme service life and does not need any internal maintenance. Only trained and competent persons should carry out maintenance.

Certain mechanical aspects of the installation should be routinely checked and rectified by a qualified fitter or electrician as below:

Monthly
Inspect for signs of wear or damage to lifting belts
Check proper operation of the winch (lift/lower)
Quarterly
Check tightness of fixing bolts and tighten if needed (ref page 7)
Check mounting structure, including welds / fasteners etc
Check for any abnormal noises
Brush down and wipe over the winch with a damp cloth

Refer to page 17 for a suitable maintenance log form

Any works requiring access to electrical connections or junction boxes must be carried out by a qualified electrician!

If the winch has exceeded the stated working life, replacement is recommended. All inspections must be monthly beyond the working life.

Disposal

If disposal is required, it should be done using the most up-to-date recycling technology according to local regulations and laws.

Drain oil via the plug provided on the gearbox, and contain it for safe disposal at a suitable recycling facility.

It is recommended that the winch be taken to a scrap metal recycling facility.

TROUBLESHOOTING

Problem	Possible Cause	Action
Winch does not raise or lower	No power	Restore power supply
	Both limit cams activated	Check / adjust limit switches
	Excess load - thermal overload tripped	Reduce load, reset / adjust overload in control
	Excess use - motor overheated	Allow to cool
	Electrical fault	Check all electrical connections
Winch does not raise	Upper limit activated	Check / adjust limit switch
	Electrical fault	Check all electrical connections
Winch does not lower	Lower limit activated	Check / adjust limit switch
	Electrical fault	Check all electrical connections
Winch makes noise	Belt misalignment	Check/adjust belt alignment as it lays on to the drum
	Motor to gearbox coupling loose	Remove winch and return to Grifco for inspection
	Internal gearbox noise	Remove winch and return to Grifco for inspection

MAINTENANCE LOG

(Copy multiples of this blank form for future record keeping)

Winch Description: (e.g. model/location)

Monthly	Date	Checked / Complete (Sign)
Inspect for signs of wear or damage to lifting belts		
Check proper operation of the winch (lift/lower)		
Monthly	Date	Checked / Complete (Sign)
Inspect for signs of wear or damage to lifting belts		
Check proper operation of the winch (lift/lower)		
Monthly	Date	Checked / Complete (Sign)
Inspect for signs of wear or damage to lifting belts		
Check proper operation of the winch (lift/lower)		
Quarterly		
Check tightness of fixing bolts and tighten if needed (ref pg 9-11)		
Check mounting structure, including welds / fasteners etc		
Check for any abnormal noises		
Brush down and wipe over the winch with a damp cloth		
Monthly	Date	Checked / Complete (Sign)
Inspect for signs of wear or damage to lifting belts		
Check proper operation of the winch (lift/lower)		
Monthly	Date	Checked / Complete (Sign)
Inspect for signs of wear or damage to lifting belts		
Check proper operation of the winch (lift/lower)		
Monthly	Date	Checked / Complete (Sign)
Inspect for signs of wear or damage to lifting belts		
Check proper operation of the winch (lift/lower)		
Quarterly		
Check tightness of fixing bolts and tighten if needed (ref pg 9-11)		
Check mounting structure, including welds / fasteners etc		
Check for any abnormal noises		
Brush down and wipe over the winch with a damp cloth		

Limited Warranty in Australia and New Zealand

1. Your consumer rights and guarantees

This Limited Warranty is provided by Chamberlain Australia Pty Ltd. Chamberlain New Zealand Limited (Chamberlain), contact details in Section 5 below. This Limited Warranty applies to a Grifco® Commercial unit (Unit) purchased in Australia and New Zealand, and gives you benefits which are in addition to your consumer rights and remedies under the Australian Consumer Law (or corresponding New Zealand consumer protection laws).

You can find out more information about your consumer rights and guarantees which the law provides in Australia at www.accc.gov.au in New Zealand at www.consumerprotection.govt.nz. We also provide this statement as required under the Australian Consumer Law:

Our goods come with guarantees that cannot be excluded under the Australian Consumer Law. You are entitled to a replacement or refund for a major failure and for compensation for any other reasonably foreseeable loss or damage. You are also entitled to have the goods repaired or replaced if the goods fail to be of acceptable quality and the failure does not amount to a major failure.

2. What does our Limited Warranty cover?

Chamberlain warrants that, when purchased new in Australia or New Zealand, the Unit (all parts of the Unit other than globes and batteries) is free from defects in materials and workmanship (Limited Warranty) for the Limited Warranty period, subject to the terms and conditions of this Limited Warranty.

The Limited Warranty period (for Accessories see below) is 2 years (24 months) from the date of purchase or when a recommended cycle count for the purchased operator has been reached (whichever comes first)

The Limited Warranty period for remote controlled transmitters and accessories included with the Unit (Accessories) is 12 months from the date of purchase.

The Limited Warranty for genuine spare parts is free from defects in material and workmanship for a period of 6 months from date of purchase.

3. Limited Warranty Conditions

The following terms and conditions apply to your Limited Warranty:

- our Limited Warranty is effective from date of purchase as indicated in Section 2 above;
- proof of purchase of the Unit is required;

Please also see the User Manual for the Unit available on our website, or provided at the time of purchase.

4. What is not covered

- Batteries and globes are not covered under the Chamberlain Limited Warranty.
- Travel costs incurred by Chamberlain or its authorised dealer in either travelling to and from areas outside a capital city area. These costs will be at the purchaser's expense.
- Additional access costs incurred by Chamberlain or its authorised dealer in obtaining access to premises where the Unit is not readily accessible. These costs will be at the purchaser's expense.

Our Limited Warranty covers defects as explained, and does not cover all problems and mishaps that may occur in relation to the Unit including:

- you got what you asked for but simply changed your mind, found it cheaper somewhere else, decided you did not like the purchase or had no use for it;
- you misused the Unit in any way that caused the problem;
- you knew of or were made aware of any deficiencies with the Unit before date of purchase;
- use of the Unit with controls or third party devices or software which has not been supplied, or pre-approved, by Chamberlain;
- problems relating to or residing in third party hardware, software or other items with which our product is used;
- any loss of data related to you or provided by you, or loss related to downtime associated with use of the product whether through power outage, failure of internet or wireless connectivity, network disruptions, or otherwise;
- non-compliance with the relevant instructions in the User Manual;
- tampering, neglect abuse, wear and tear, accident, electrical storm, excessive use or conditions other than normal use;

- problems with, or relating to, hardware, including but not limited to the springs, hinges, guides;
- problems caused by electrical faults or replacement of batteries or light bulbs, blown fuses, electrical surges, power surges or power strikes, fire, flood, rain, water, lightning or storms;
- water or moisture ingress that causes corrosion or electrical malfunction; corrosion caused by sea air if located near a waterway, beach etc: lack of proper maintenance, service or care of the Unit;
- and/or damage caused by insects, pests or other after sale damage caused by events or accidents outside Chamberlain's reasonable control and not arising under normal and standard operating conditions.

5. Where you need help with our product

If you have a problem with the Unit or if you are concerned it may be defective, please contact our Customer Service team below:

Australia	Phone toll free 1800 474 326
Fax toll free	1800888121
Address	Unit 1, 75 Epping Road North Ryde, NSW 2113
Email	customerservice@chamberlainanz.com
Website	https://www.grifco.com.au/
New Zealand	Phone toll free 0800 653 667
Auckland	Phone 09 477 2823
Fax toll free	0800653 663
Email	customerservice@chamberlainanz.com
Website	https://www.grifco.co.nz/

As a first step, our Customer Service team will provide product support assistance to help you try to resolve the problem and in some cases, replacement parts for "do it yourself repairs". If our technical team need to see the product to determine the nature of the fault we will arrange an inspection by a trained technician, or ask that you send the product to us.

If your product was installed at your premises by an authorised dealer, please contact the installer indicated in your sales documents for prompt on-site service, or consult our Customer Service team if you need help in finding a local service technician. A service fee for on-site service may apply.

Authorisation from Chamberlain is required for warranty service under this Limited Warranty.

6. Delivery Charges

If an inspection is required in order to determine the nature of the fault please send the Unit freight paid and well packaged in accordance with our instructions. Any costs associated with transporting the Unit will be at Chamberlain's cost if the Unit has breached a statutory consumer guarantee. Repairs and replacement parts provided under this Limited Warranty are free of charge (and repaired items or replacement parts for Units found to be defective will be returned to you at our cost), and warranted by Chamberlain for the remaining portion of the original warranty period. However, if you make a claim under this Limited Warranty and this Limited Warranty does not apply, or if the law does not provide you with a statutory right or remedy as a consumer, you may be required to pay labour, assessment and/or freight costs and delivery charges to repair, replace and/or return the Unit to you.

7. How to register your product

Please register your Unit and contact details at this link <https://www.grifco.com.au/warranty-registration> so that we or your authorised installer can locate your details and provide you with product support assistance as quickly as possible, and keep you up to date with product updates. Registration can also be done by contacting Chamberlain Customer Service.

8. Important Reminder

All other guarantees (other than your statutory rights and guarantees as a consumer under applicable laws) are excluded to the fullest extent that we may lawfully do so. Unless the law requires us to do so, we do not accept any other liability, such as compensation for damages, injury or loss.

Last updated; 21st April 2022